DESIGN ASSESSMENT

Propose a Frequency Allocation Strategy for Mixed Wireless Services (BL6)

A priority-based frequency allocation framework for heterogeneous wireless services with guard-band protection, interference checking, dynamic reallocation and MATLAB simulation.

1. Aim

To design and propose a frequency allocation strategy for mixed wireless services that efficiently assigns the available spectrum to different wireless services according to their bandwidth requirements and priority levels while minimizing interference and improving spectrum utilization.

2. Objectives

- Identify the spectrum requirements of different wireless services.
- Divide the available spectrum into suitable frequency channels.
- Assign priority levels to different wireless services.
- Allocate frequency bands according to service requirements.
- Provide guard bands between neighboring services.
- Minimize adjacent-channel interference.
- Dynamically reallocate unused spectrum whenever required.
- Improve overall spectrum utilization.

3. Services Considered

Wireless Service	Priority	Requirement
Emergency/Public Safety	5	Medium
5G/6G Cellular	4	High
V2X Communication	3	Medium
IoT Communication	2	Low
Wi-Fi Communication	1	Medium

4. Requirements

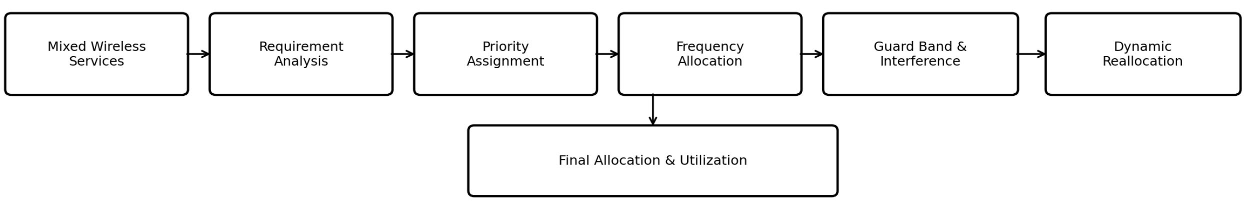
Hardware: Computer/Laptop; minimum 4 GB RAM; standard processor; display.

Software: MATLAB / MATLAB Online; MATLAB Communication Toolbox (optional); Windows/Linux.

Programming: MATLAB; frequency-channel generation; priority allocation; bandwidth, guard-band, interference and utilization calculations.

5. System Concept

Basic operation: Available Spectrum → Service Request → Priority Assignment → Frequency Allocation → Guard Band → Interference Check → Dynamic Reallocation → Spectrum Utilization.



6. Proposed Spectrum Allocation

An illustrative spectrum range of 3.4 GHz–3.5 GHz is considered, giving 100 MHz total bandwidth. This is a proposed assessment design, not a regulatory licensed-band assignment.

Frequency Range	BW	Assigned Service	Priority
3400–3410 MHz	10 MHz	Emergency/Public Safety	5
3410–3415 MHz	5 MHz	Guard Band	—
3415–3450 MHz	35 MHz	5G/6G Cellular	4
3450–3455 MHz	5 MHz	Guard Band	—
3455–3470 MHz	15 MHz	V2X Communication	3
3470–3475 MHz	5 MHz	Guard Band	—
3475–3485 MHz	10 MHz	IoT Communication	2
3485–3490 MHz	5 MHz	Guard Band	—
3490–3500 MHz	10 MHz	Wi-Fi Communication	1

7. Bandwidth Verification

Service bandwidth = 10 + 35 + 15 + 10 + 10 = **80 MHz**
Guard bandwidth = 5 × 4 = **20 MHz**
Total spectrum = 80 + 20 = **100 MHz**
Spectrum utilization = (80/100) × 100 = **80%**.

8. Working Principle

1. Initialize the 3.4–3.5 GHz spectrum.
2. Identify services requesting spectrum.
3. Assign service priorities.
4. Determine bandwidth requirements.
5. Allocate spectrum according to priority and bandwidth.
6. Insert guard bands.
7. Check for frequency overlap/interference.
8. Dynamically reallocate when a higher-priority request occurs.
9. Calculate spectrum utilization.

9. Algorithm

1. Start; define available spectrum.
2. Define services, priorities and bandwidths.
3. Allocate the highest-priority service first.
4. Insert a guard band after each allocation except the last.
5. Repeat allocation for remaining services.
6. Check frequency overlap and shift an affected channel if necessary.
7. Calculate allocated bandwidth and spectrum utilization.
8. Display the allocation table and plot the result.
9. Stop.

10. MATLAB Code

```
clc;
clear;
close all;

% Total spectrum: 3.4 GHz to 3.5 GHz
start_freq = 3400;       % MHz
end_freq   = 3500;       % MHz

% Wireless services
services = {'Emergency', '5G/6G', 'V2X', 'IoT', 'Wi-Fi'};

% Priority values
priority = [5 4 3 2 1];

% Required bandwidth in MHz
bandwidth = [10 35 15 10 10];

% Guard band in MHz
guard_band = 5;

% Number of services
N = length(services);

% Storage for allocated frequencies
start_alloc = zeros(1,N);
end_alloc   = zeros(1,N);

% Current frequency
current_freq = start_freq;

% Frequency allocation
for i = 1:N
    start_alloc(i) = current_freq;
    end_alloc(i) = current_freq + bandwidth(i);
    current_freq = end_alloc(i);

    % Add guard band except after the last service
    if i < N
        current_freq = current_freq + guard_band;
    end
end

% Display allocation table
fprintf('Frequency Allocation Strategy\n');
fprintf('--------------------------------------------\n');

for i = 1:N
    fprintf('%-12s : %6.0f - %6.0f MHz\n', ...
        services{i}, start_alloc(i), end_alloc(i));
end

% Calculate service bandwidth
service_bandwidth = sum(bandwidth);

% Calculate guard-band bandwidth
total_guard_band = guard_band * (N-1);

% Calculate spectrum utilization
utilization = (service_bandwidth / (end_freq-start_freq)) * 100;

fprintf('\nTotal Available Spectrum = %.0f MHz\n', ...
    end_freq-start_freq);
fprintf('Service Bandwidth = %.0f MHz\n', service_bandwidth);
fprintf('Guard Bandwidth = %.0f MHz\n', total_guard_band);
fprintf('Spectrum Utilization = %.2f %%\n', utilization);

% Plot frequency allocation
figure;
hold on;
for i = 1:N
    plot([start_alloc(i) end_alloc(i)], [i i], 'LineWidth', 8);
    text((start_alloc(i)+end_alloc(i))/2, i, services{i}, ...
         'HorizontalAlignment', 'center', ...
         'VerticalAlignment', 'middle');
end

xlabel('Frequency (MHz)');
ylabel('Wireless Service');
title('Proposed Frequency Allocation for Mixed Wireless Services');
yticks(1:N);
yticklabels(services);
xlim([start_freq end_freq]);
grid on;
hold off;
```

11. Expected Output

The program produces a Command-Window allocation table, a frequency allocation graph, and the calculated spectrum utilization.

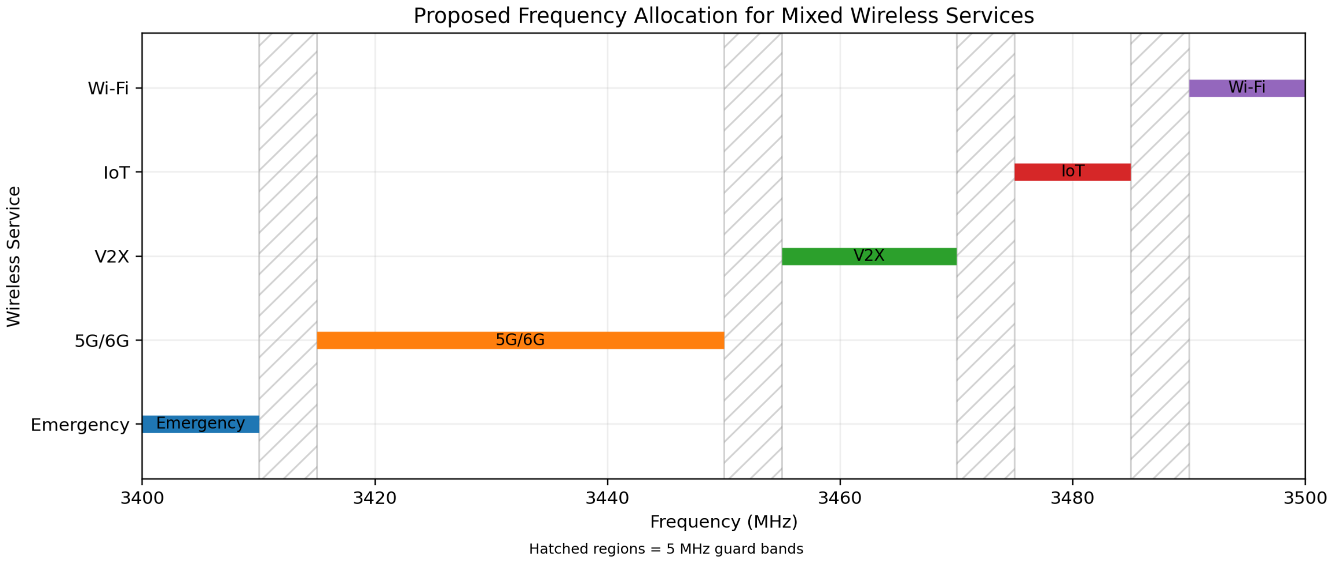
12. MATLAB Command-Window Output

The following is the exact deterministic output corresponding to the `fprintf` statements in the MATLAB code.

```
Frequency Allocation Strategy
---------------------------------------------
Emergency   :   3400 -   3410 MHz
5G/6G       :   3415 -   3450 MHz
V2X         :   3455 -   3470 MHz
IoT         :   3475 -   3485 MHz
Wi-Fi       :   3490 -   3500 MHz

Total Available Spectrum = 100 MHz
Service Bandwidth = 80 MHz
Guard Bandwidth = 20 MHz
Spectrum Utilization = 80.00 %
```

13. MATLAB-Style Graphical Output



The graph uses the exact frequency intervals calculated by the MATLAB code. Hatched regions represent the 5 MHz guard bands.

14. Result Verification

Parameter	Verified Value
Total available spectrum	100 MHz
Emergency	3400–3410 MHz
5G/6G	3415–3450 MHz
V2X	3455–3470 MHz
IoT	3475–3485 MHz
Wi-Fi	3490–3500 MHz
Service bandwidth	80 MHz
Guard bandwidth	20 MHz
Spectrum utilization	80.00 %

15. Interference Management

- **Frequency separation:** separate frequency ranges are assigned to different services.
- **Guard bands:** 5 MHz is placed between neighboring services.
- **Priority-based reallocation:** available or lower-priority temporary spectrum can be reassigned to a higher-priority service.

16. Advantages

- Efficient allocation of limited spectrum.

- Supports multiple wireless services simultaneously.
- Reduces adjacent-channel interference.
- Prioritizes critical communication services.
- Guard bands improve frequency isolation.
- Supports dynamic spectrum reallocation.
- Improves spectrum utilization.
- Extendable to 5G/6G.
- Suitable for dense wireless environments.
- Flexible for heterogeneous wireless networks.

17. Applications

- 5G/6G cellular networks
- Smart cities
- IoT networks
- Wi-Fi networks
- Vehicular communication
- Emergency/public-safety communication
- Industrial wireless networks
- Dense urban communication systems
- Cognitive radio networks
- Intelligent spectrum management

18. Result

Thus, a frequency allocation strategy for mixed wireless services was successfully proposed. A 100 MHz spectrum range from 3.4 GHz to 3.5 GHz was divided among Emergency/Public Safety, 5G/6G, V2X, IoT and Wi-Fi services based on priority and bandwidth requirements. Guard bands were introduced between neighboring services to minimize adjacent-channel interference. The allocation uses 80 MHz for wireless services and 20 MHz for guard bands, resulting in 80% spectrum utilization. The strategy provides a flexible priority-based approach for managing spectrum among heterogeneous wireless services while improving spectrum efficiency and reducing interference.